

OS-Level Omnipresent Copilot Framework

CRM Research Overlay + Design System Pattern Templates

Audience: Chief Design Officer review & next-week brainstorming

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North-star: The intent bar routes. The sidebar works. Inline assists. The rail nudges.

Executive summary

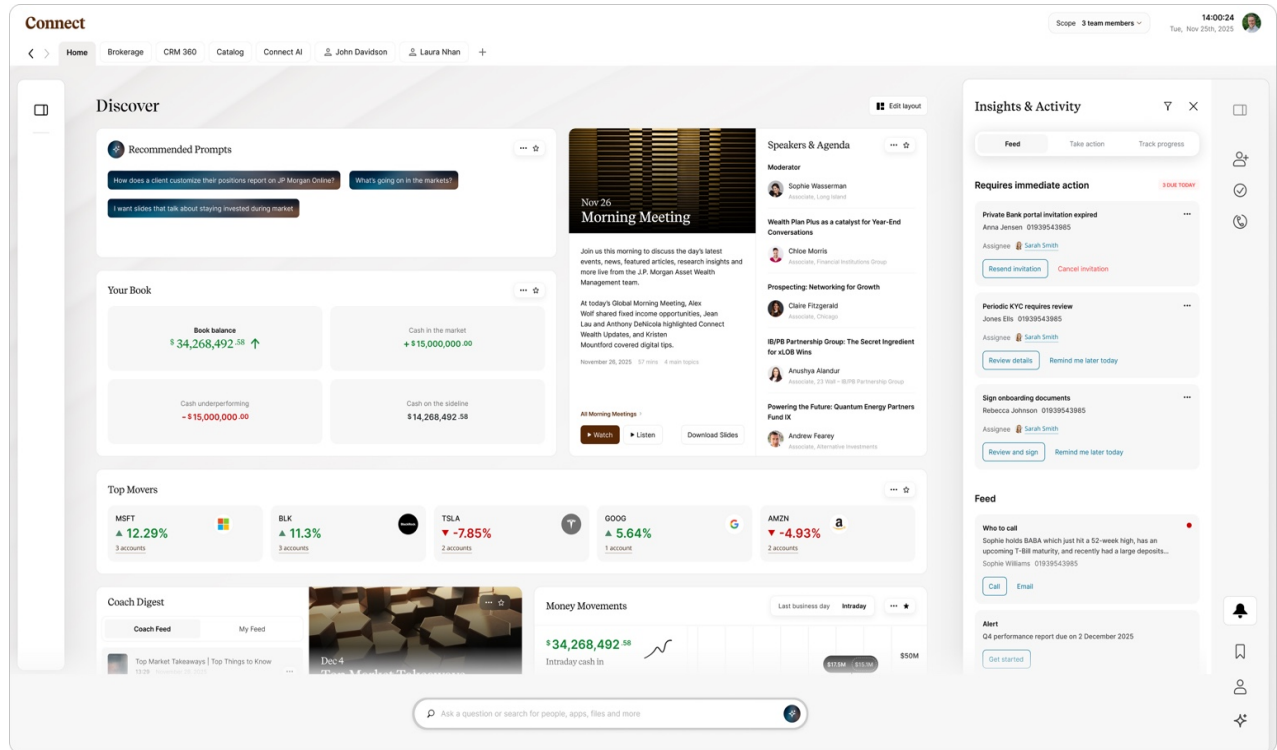
- We're designing an OS-level copilot that stays consistent across apps while respecting regulated-work guardrails.
- CRM research indicates advisors spend the majority of time on administrative work and must manually compile client intelligence across 10+ disconnected systems.
- This environment demands: (1) a global, low-friction entry for ambiguous intent, and (2) a contextual workspace for multi-step reasoning, artifacts, and approvals.
- The resulting architecture is a hybrid: global intent bar as the router + right sidebar as the workspace, reinforced by inline micro-assists and proactive rails.

Research-informed product requirements (overlay)

- **Attention is stretched thin** → Copilot must prioritize and surface "what needs attention now" with clear urgency + impact signals; default to proactive rail + daily digest.
- **Information is fragmented across systems** → Copilot must offer OS-level search/routing and attach explicit context (record/app) for synthesis and action.
- **Process coordination is opaque** → Copilot must model work items (status, blockers, owners, SLAs) and provide a "plan + handoff" pattern when automation is partial.
- **Market events must become client-specific talking points** → Copilot needs a standard pattern for "event → implications → talking points → next steps" with provenance.
- **Trust & compliance matter** → Copilot must show scope, sources/provenance, and require confirmations for external communications and transactions.

Current UI reference (for discussion)

This screenshot illustrates an existing right-rail activity feed (alerts + quick actions) alongside a central discovery canvas. The framework below formalizes this split: center/global for intent routing and exploration; right for action, evidence, and follow-through.



1. Core model: Locality × Depth

Every copilot request is classified on two axes: (A) where context lives, and (B) how deep the interaction needs to be. These axes select the leading surface and pattern.

Axis A: Context locality

- **Global (OS-level):** cross-app, cross-client. Example: “Who should I talk to right now?”
- **App:** within one app’s state. Example: “What routine maintenance is outstanding here?”
- **Record:** tied to a client/account/case/doc. Example: “What changed for this client?”
- **Workflow/Form:** a transactional step. Example: “Help me complete this trade ticket.”

Axis B: Interaction depth

- **0–1 step:** lookup, navigate, quick answer.
- **2–4 steps:** refine, compare, choose, preview.
- **Multi-step / transactional:** creates/updates/sends/books/submits; needs validation + approval.

Rule of thumb: Locality decides where context comes from; depth decides how much workspace is required. Global + shallow → intent bar. Record + deep → sidebar. Workflow + deep → inline + sidebar.

2. Canonical copilot surfaces (and why each exists)

Research points to both high context-switching and high admin load. To reduce friction, each surface has a clear job.

Surface	Role	Best for	Research linkage
Omnipresent Intent Bar	Start + route + find + navigate	Global intent, entity search, app switching, quick answers	Counters fragmentation across 10+ systems by providing one OS-level entry
Right Sidebar (Workspace)	Multi-turn reasoning + artifacts + audit trail	Synthesis, previews, approvals, guided flows, history, provenance	Supports multi-step coordination + makes trust controls visible
Inline Assist	Micro-assistance at point of work	Prefill, explain, validate during forms/transactions	Reduces admin effort inside high-cost workflows
Proactive Rail / Feed	Ambient intelligence + prioritization	Urgent tasks/risks/opps with quick actions	Direct response to “no integrated system that surfaces what needs attention now”

3. Locality × Depth mapping matrix

This is the primary decision tool for design reviews.

Locality \ Depth	0–1 step (lookup/answer)	2–4 steps (refine/decide)	Multi-step / transactional
Global (OS-level)	Intent bar: command palette / quick answer. Output: results, shortcuts.	Bar → Sidebar: deep answer (synthesis). Output: options, sources, next steps.	Sidebar: guided action or plan + handoff. Output: stepper, approvals.
App	Intent bar scoped to app: quick answer/palette.	Sidebar: compare, summarize, recommend within app context.	Sidebar + Inline: guided action; inline if form-based.
Record	Sidebar: “catch me up” (pinned record copilot).	Sidebar: deep answer + evidence + recommended actions.	Sidebar + Inline: execute with record context + strict gates.
Workflow / Form	Inline: explain / prefill / validate.	Inline + Sidebar: refine details; preview; resolve exceptions.	Inline + Sidebar: confirm, execute, document (audit).

Escalation invariant: if a request becomes multi-turn, requires provenance, or changes the world (send/submit/book/trade), continue in the sidebar.

4. CRM research overlay: use-case categories → patterns

The CRM research defines six deep-research categories. Below is how each maps to our surfaces and patterns.

Research category	What advisors need	Default surface(s)	Pattern(s)
Client Intelligence & Profiling	360° view, life-change detection, relationship context, provenance	Sidebar (record) + Intent bar (global entry)	Record Copilot (Pinned), Deep Answer (Synthesis)
Opportunity & Risk Identification	Early warnings, drift signals, ranked opportunities, options with pros/cons	Rail (proactive) + Sidebar	Proactive Card, Deep Answer, Guided Action Flow
Content & Engagement Personalization	Hyper-personalized thought leadership + talking points; review before send	Sidebar + Intent bar	Deep Answer, Guided Action (draft/send) with confirmation
Prospecting & Lead Qualification	Warm paths, batch research, engagement drafts with human review	Sidebar + Rail (proactive)	Deep Answer, Guided Action (multi-touch drafts) + Audit
Internal Coordination & Handoffs	Process status, SLAs, blockers, routing to CSS/teams, expectation mgmt	Sidebar	Plan + Handoff, Guided Action Flow, Work-item tracker
Market & Sector Intelligence	Event → client implications → talking points at scale	Rail + Sidebar	Proactive Card, Deep Answer (talking points)

5. Design system pattern templates

The following templates are formatted for design-system governance: each pattern includes intent, triggers, scope, components, states, content rules, guardrails, and telemetry.

Command Palette Results

- **Intent:** Find/open entities or apps quickly across the suite.
- **Entry points:** OS shortcut (Cmd/Ctrl+K), click intent bar, slash commands.
- **Scope:** Defaults to All Apps; user can narrow via scope chips.
- **Core components:** IntentBar, ScopeChips, ResultsList, EntityRow, QuickActions.
- **States:** idle, focused, querying, results, empty, error.
- **Escalation:** If user requests analysis (“why”, “compare”, “summarize”), continue in Sidebar Deep Answer.
- **Content rules:** Use concise labels; show entity type + last-updated where possible.
- **Telemetry:** query→click-through, time-to-open, refinement rate, no-result rate.

Quick Answer Overlay

- **Intent:** Provide a 0–1 step answer without leaving flow.
- **Entry points:** Intent bar question; keyboard submit.
- **Scope:** Must display scope chip inline in the answer.
- **Core components:** AnswerCard, SourcePills (optional), ExpandToSidebar CTA.
- **States:** loading, answer, partial-answer (access-limited), error.
- **Guardrails:** No execution. No external comms. Provide “View details” for evidence.
- **Telemetry:** dwell time, expand rate, follow-up rate, satisfaction micro-feedback.

Deep Answer (Synthesis)

- **Intent:** Cross-system synthesis with provenance and follow-ups.
- **Entry points:** From intent bar (auto-escalate) or sparkles/sidebar.
- **Scope:** Explicitly show data sources + timestamps; indicate access constraints.
- **Core components:** SidebarWorkspace, ThreadHeader (context chips), AnswerSections, EvidenceDrawer, RecommendedActions.
- **States:** thinking/plan, drafting, final, cited, refresh, error.
- **Guardrails:** Explainability + provenance; avoid hallucinated facts; encourage verification for high-stakes.
- **Telemetry:** action adoption, citation opens, refreshes, time-to-decision.

Guided Action Flow

- **Intent:** Turn an intent into a safe, reviewed execution (send/book/submit/trade).
- **Entry points:** From sidebar recommendation; from proactive card “Act”.
- **Scope:** Record/app context locked; show what will be changed before execution.
- **Core components:** Stepper, PreviewPanel, ValidationChecks, ConfirmationGate, AuditSummary.
- **States:** gather inputs, preview, confirm, executing, success, failure, rollback guidance.
- **Guardrails:** Human-in-loop required; compliance disclaimers; block if missing approvals.
- **Telemetry:** completion rate, abort points, validation failures, time per step.

Record Copilot (Pinned)

- **Intent:** Always-on record intelligence: “catch up”, risks/opps, next best actions.
- **Entry points:** Open sidebar while on a record; pin toggle persists across navigation.
- **Scope:** Single record; show last-refresh time and included sources.
- **Core components:** RecordHeader, CatchUpSummary, SignalsTimeline, NextActions, Notes/Memory (policy).
- **States:** pinned/unpinned; record-switch; stale; refreshed.
- **Guardrails:** Never infer outside access; show “incomplete due to permissions”.
- **Telemetry:** pin usage, catch-up frequency, action click-through.

Inline Assist (Form Micro-actions)

- **Intent:** Reduce admin work at the point of transaction without breaking flow.
- **Entry points:** Inline sparkle, “prefill with AI”, “explain field”.
- **Scope:** Workflow step only; anything broader escalates to sidebar.
- **Core components:** InlineSuggestion, PrefillButton, ExplainPopover, ValidationHint.
- **States:** suggestion, accepted, rejected, edited, escalated.
- **Guardrails:** Don’t auto-submit; always require user review; log changes.
- **Telemetry:** accept/reject rates, edit distance, time saved proxy.

Proactive Card (Explain / Act / Snooze)

- **Intent:** Surface “what needs attention now” in priority order with minimal noise.
- **Entry points:** Right rail/feed; daily digest page.
- **Scope:** Show why the card exists (signal) + confidence + impact.
- **Core components:** AlertCard, ExplainCTA (opens sidebar), ActCTA (guided flow), SnoozeControl.
- **States:** new, viewed, acted, snoozed, resolved.
- **Guardrails:** Proactive only for high-confidence items; avoid alert fatigue.
- **Telemetry:** act rate, snooze rate, false-positive feedback, time-to-resolution.

Plan + Handoff (No Infra)

- **Intent:** Provide value even when systems can't be automated yet.
- **Entry points:** Any guided action that requires manual steps or another tool.
- **Scope:** List the systems involved; provide deep links + checklist.
- **Core components:** PlanCard, Checklist, DeepLinks, StatusNotes, FollowUpReminders.
- **States:** planned, in-progress, blocked, completed.
- **Guardrails:** Be explicit about what copilot can/can't do; never claim execution.
- **Telemetry:** handoff completion, link usage, drop-off reasons.

6. Governance, measurement, and CDO discussion prompts

Use this section to drive next week's brainstorming toward crisp decisions.

Governance model (design system + product)

- Treat each pattern as a governed spec: components, states, copy rules, and required trust UI (scope + provenance).
- Define a "graduation" rule: bar/overlay may not execute; execution always happens via guided flow with confirmation.
- Create an intent taxonomy (Find / Ask / Do / Monitor) to avoid the bar becoming a junk drawer.
- Establish evaluation gates for new skills: confidence thresholds, access boundaries, and rollout tiers.

Measurement (what success looks like)

- Time saved proxies: reduced clicks, reduced context switching, fewer app hops per job.
- Attention outcomes: proactive cards acted vs snoozed; noise/false-positive feedback.
- Business outcomes: opportunities surfaced, meeting prep time reduction, response latency on urgent tasks.
- Trust outcomes: evidence opens, overrides/edits, compliance exceptions, user confidence ratings.

CDO brainstorming prompts (decisions to make)

- What is the default scope for the OS-level bar: All Apps, My Book, or Active Window?
- Should the right rail and sidebar be separate surfaces or tabbed within a single right dock?
- What is the minimum provenance UI for synthesis answers: source pills, evidence drawer, or full citations?
- How should pinning work across apps: per-thread, per-record, or global session?
- What proactive items are allowed by policy: only compliance + urgent ops, or also growth opportunities?
- What is our "human-in-loop" standard for drafts (email/content) and transactions (trade/approvals)?

References

AI_Native_Deep_Research_Products for Wealth Management: Strategic Analysis & Product Blueprint (provided by team).